Sascha Friederichs

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Birthday and -location	18.02.1976 in Heinsberg
Nationality	German



Occupation

April 2026 – now

Freelance cloud & systems architect (self-employed)

- Designed and deployed a serverless AWS inventory application using Vite and AWS S3 for the web front-end, alongside AWS API Gateway, Lambda, and DynamoDB for the back-end; infrastructure provisioned via Terraform (IaC) with automated CI/CD pipelines through GitHub and GitHub Actions
- Designed a static AWS cloud portfolio website using Vite and GitHub (CI/CD), leveraging AWS Amplify for hosting, Route 53 for DNS management, and DynamoDB for a global visitor counter
- Prepared and successfully achieved the AWS Certified Solutions Architect – Associate certification (August 2026)

July 2025 – March 2026

Lead system architect at Aumovio Autonomous Mobility Germany GmbH, Eschborn

- Technical support for customer acquisition and representative for systems engineering in customer meetings
- Development of a leaner systems engineering process
- Presenter in Aumovio's System Engineering Conference 2025, topic: "Evaluating Catia Magic as an Alternative to IBM Rhapsody for System Architecture Modeling"
- Development of Rhapsody AI plugins for model checks
- Designed scalable system architecture and requirements for autonomous driving trucks

July 2022 – June 2025

Lead system architect at Continental Autonomous Mobility Germany GmbH, Eschborn

- Technical leadership for the ADAS system engineering team
- Support and training for system modeling language (SysML), IBM Rhapsody and the IBM Harmony method
- Setup of system architecture models for autonomous and assisted driving control units (ADCU)
- Review of system architecture models and design requirements for product releases
- Development of concepts for platform functions e. g. over the air update (OTA) or system state machine

November 2019 – June 2022

System architect at Continental Teves AG & Co. oHG, Eschborn

- Rational Harmony method with SysML and IBM Rational Rhapsody for autonomous and assisted driving control units (ADCUs) in customer projects
- System architecture design according to IBM

- Creation of system requirements from customer and safety specifications in IBM Rational Doors for ADCUs
- Development of ConOps for ADCUs in MS-Powerpoint

February 2019 – October 2019 Freelancer software developer and software architect

- Development of Android applications with Java
- Development of web-based applications with HTML, CSS and Java
- Use of Google speech-to-text API
- Monetization with Google AdMob/AdSense integration

July 2017 – January 2019 Supervisor system architecture at ZF Active Safety GmbH, Koblenz

- Technical and disciplinary lead for a team of five system architects
- Responsible for the system architecture of two camera projects with an annual turnaround of > 250 million EUR
- Successful support of development in the role of senior system architect with four international development locations during the entire development phase
- Responsible for quality and assuring of the same by reviews and international workshops
- Planning of resource usage and timing for system architecture artifacts with PTC Integrity

May 2016 – June 2017 System architect at ZF Active Safety GmbH, Koblenz

- System architecture design according to OOSEM with SysML in Rhapsody for driver assistance camera systems
- Creation of ConOp documents in MS-Powerpoint
- Writing of software requirements in Doors
- Development of methods for the Automotive-SPICE-Process for Eng.3 and Eng.4 levels
- Coordination with customers and debugging

November 2013 – April 2016 Software architect at ZF Active Safety GmbH, Koblenz

- Design of software architecture with UML in Rhapsody for driver assistance systems and data fusion ECUs
- Development of the AUTOSAR models for camera systems with AutosarBuilder
- Introduction of C code generation from Rhapsody
- Writing of software requirements in Doors

November 2011 – October 2013 Software architect at ZF Lenksysteme GmbH, Schwäbisch Gmünd

- Development of software architecture models for steering systems in UML with EnterpriseArchitect
- Execution of safety analysis according to ISO26262
- Test and evaluation of development tool AutosarBuilder for global introduction as tool for AUTOSAR modeling at ZF Lenksysteme GmbH

November 2008 – October 2011 Project manager at ZF Lenksysteme GmbH, Shanghai

- Technical lead for a team of six engineers
- Coordination of simultaneous engineering with mainly Japanese and Chinese suppliers as well as the German headquarter
- Product presentations at Chinese, Japanese and Malayan customers
- Planning, coordination and execution of customer acquisition
- Planning and supervision of demonstration vehicle buildup in own facilities and customer locations

November 2006 – October 2008 Software developer at ZF Lenksysteme GmbH, Schwäbisch Gmünd

- Development of software components for electric steering systems in C with Eclipse

- Execution of software integration for delivery of software releases to customers
- Execution of software module-, MISRA-, Polyspace- and integration tests

August 2004 – October 2006 Development engineer for hardware and software at Gigatronik GmbH, Stuttgart

- Research project for protection 42V vehicle electric systems
- Calibration and initial operation of diesel engine ECUs on engine test benches and in vehicles with XCP
- Development of a device for simulation of the cam- crankshaft system of combustion engines for testing of engine ECUs

Education

March 2003 – July 2004 Postgraduate studies for MSc in Modeling Simulation and Control at Coventry University, Coventry

- Overall grade: 68% (1,7)
- Thesis grade: 70% (1,3)

September 1999 – February 2003 Undergraduate studies for Dipl. Ing. (FH) Electrical Engineering focus Automation at University of Applied Sciences Aachen, Jülich

- Overall grade: 2,2
- Thesis grade: 1,3

Skills

Certificates

- AWS Certified Solutions Architect (Associate Level)
- iSAQB Certified Professional for Software Architecture (Foundation Level)
- OMG-Certified UML Professional (Intermediate Level)
- OMG-Certified Systems Modeling Professional (Model Builder)
- Oracle Certified Associate, Java SE 7 Programmer
- Chinese Proficiency Test, HSK (Level 3)

Cloud & DevOps

- Cloud Infrastructure: AWS Cloud (Core Services, API Gateway, Lambda, S3, DynamoDB, Amplify, Route 53, AWS Certificate Manager)
- Frameworks & Methodologies: AWS Well-Architected Framework (Proficient Badge)
- Infrastructure as Code: Terraform
- Automation & CI/CD: GitHub Actions, Git, DevOps Pipelines
- Environments & Scripting: Linux, Python, Bash, Java

Systems Engineering & Architecture

- Methodologies: Arc42, SysML, UML, OOSEM, System State Machines
- Standards & Compliance: ISO 26262 (Functional Safety), ISO 21434 (Cybersecurity), ASPICE, MISRA
- Modeling Tools: IBM Rhapsody, Catia Magic, EnterpriseArchitect, IBM Rational Doors

Languages & Projects

- Languages: German (mother tongue), English (business fluent), Chinese (B1 level / HSK Level 3)
- Automotive Series Projects: Audi (HCP2, MLBevo), BMW (KaFAS4), VW (MQB), Daimler (MCM), Aurora, Tata, Honda (HA33), GM (EOCM2)

Bendorf, 10th October 2026

